Dynamic Global Illumination

2026.10.23

허준

목 차

1. 목표와 배경

2. Infra 구축

3. Infra 검증

4. 진행 계획

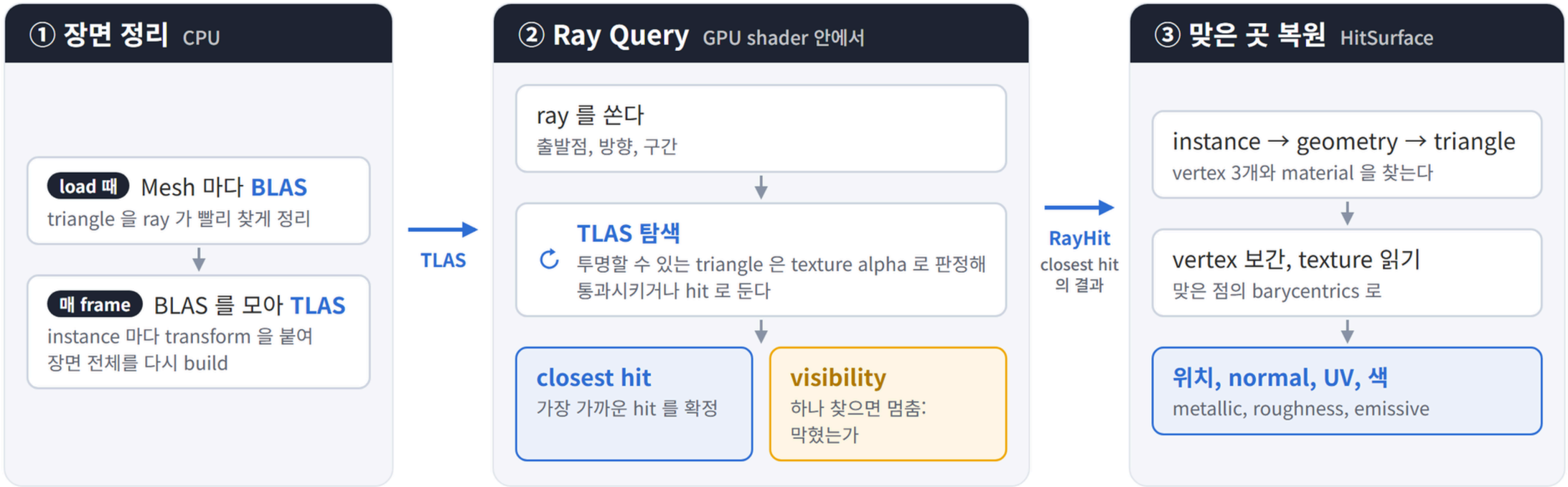
목표와 배경

임의의 위치에 ray를 쏴서 맞춘 곳의 정보를 알 수 있어야 한다

목표와 배경

Ray Query로 가져오는 핵심 정보 세 가지

Infra 구축



- BLAS: Mesh 단위로 만들어지는 BVH
- TLAS: BLAS instance로 구성된 scene BVH

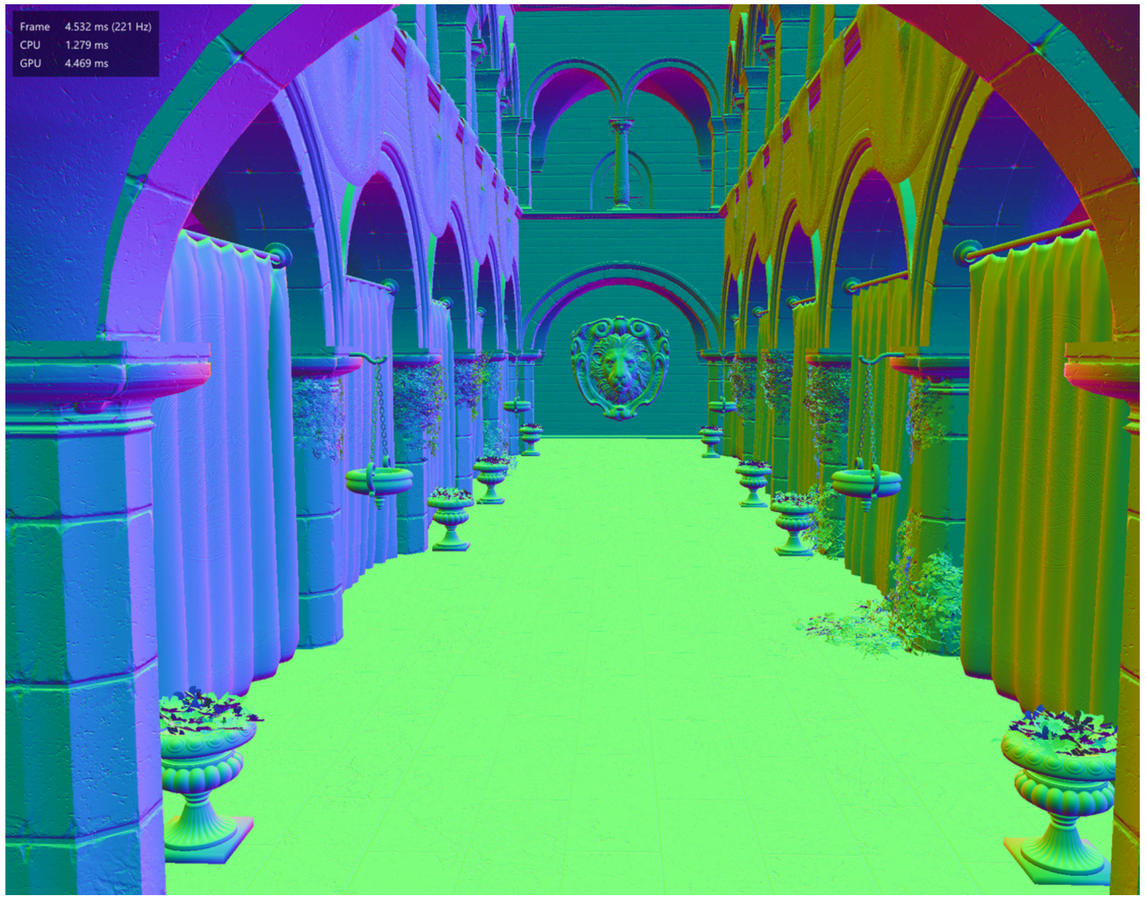
Infra 구축

Scene View

Debug View – Hit Distance

Infra 구축

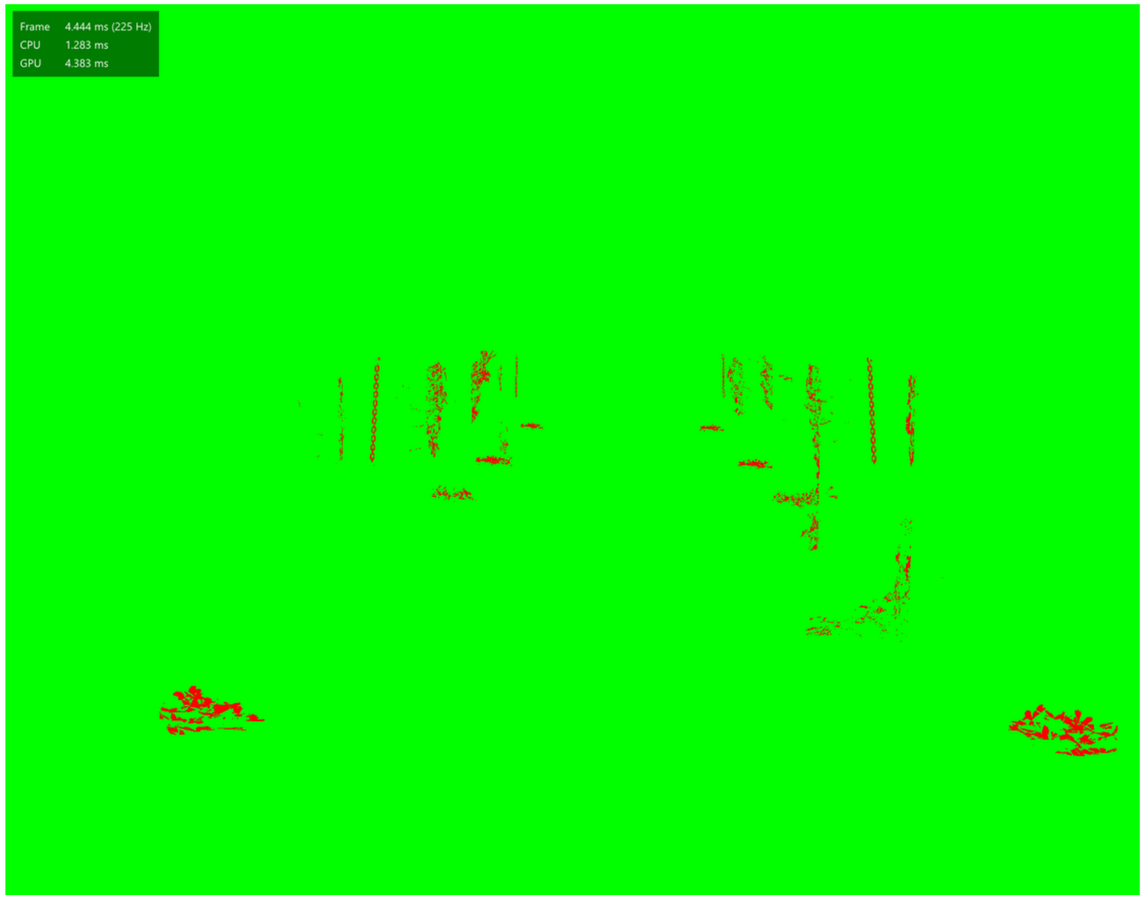
Scene View



Debug View – Normal

Infra 구축

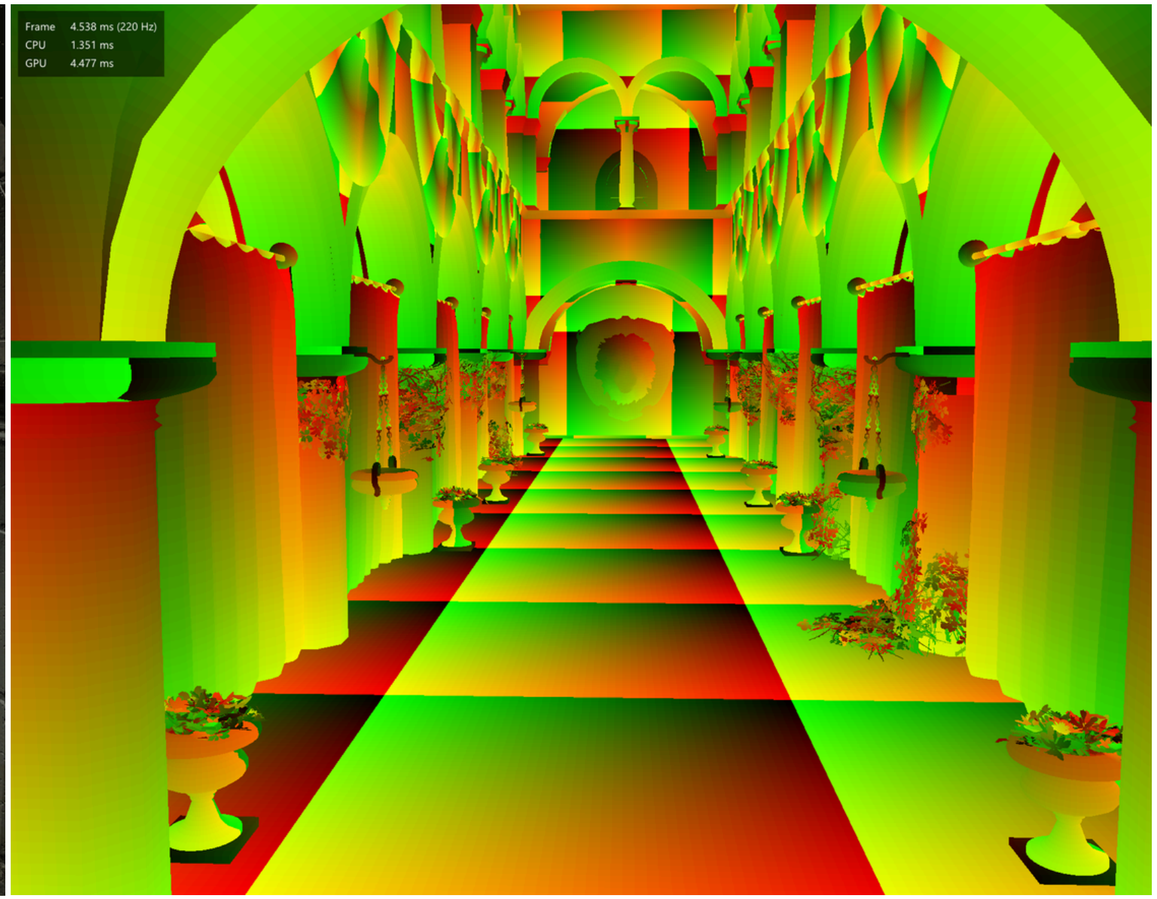
Scene View



Debug View – Front Face

Infra 구축

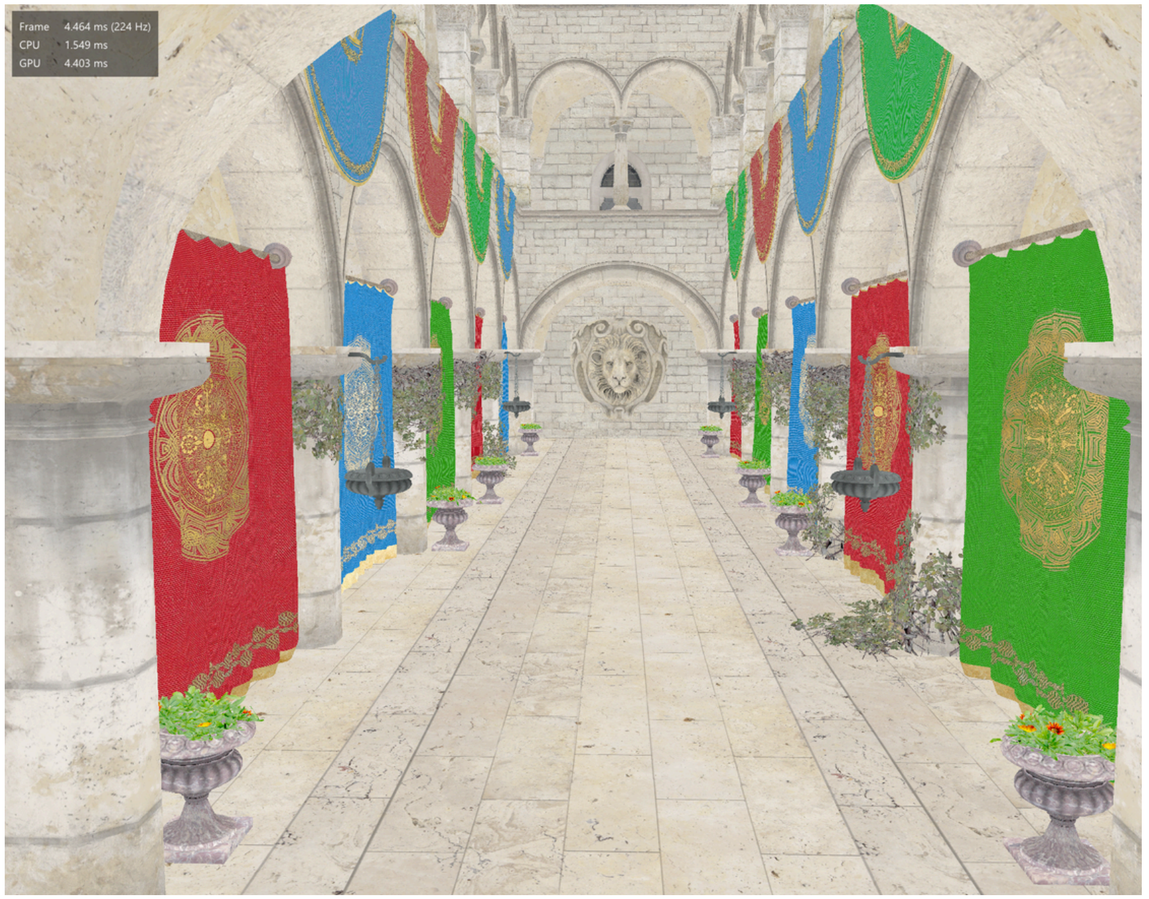
Scene View



Debug View – UV0

Infra 구축

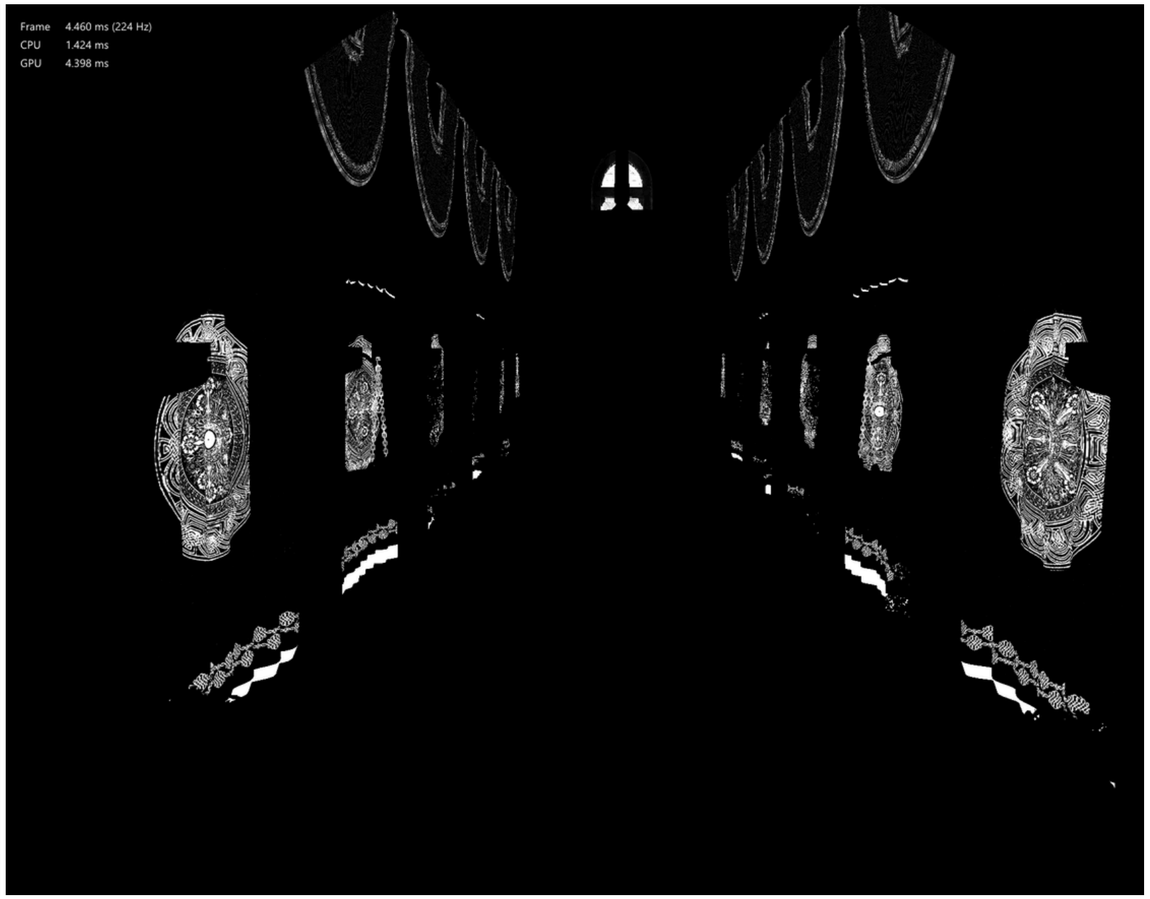
Scene View



Debug View – Base Color

Infra 구축

Scene View



Debug View – Metallic

Infra 구축

Scene View

Debug View – Roughness

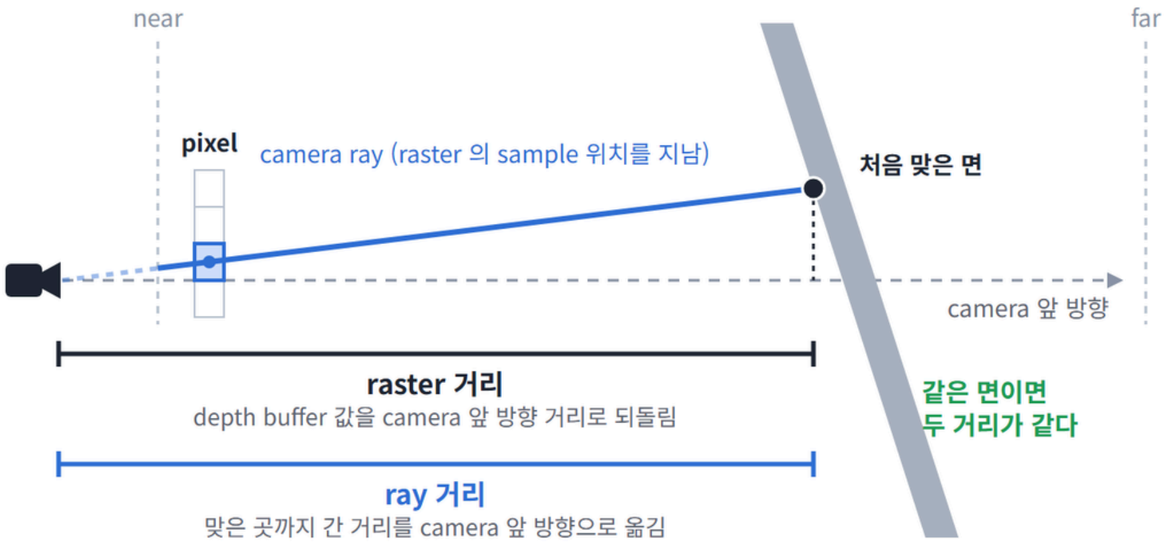
Infra 구축

Scene View

Debug View – Sun Visibility

Infra 검증

① 같은 pixel 에서 두 거리를 잰다



② 비교

상대 차이 = 두 거리의 차이 ÷ **raster 거리**

✅ **기준 이하면 같은 면을 봤다**

거리에 대한 비율로 재서 가까운 곳과 먼 곳을 기준 하나로 판정한다

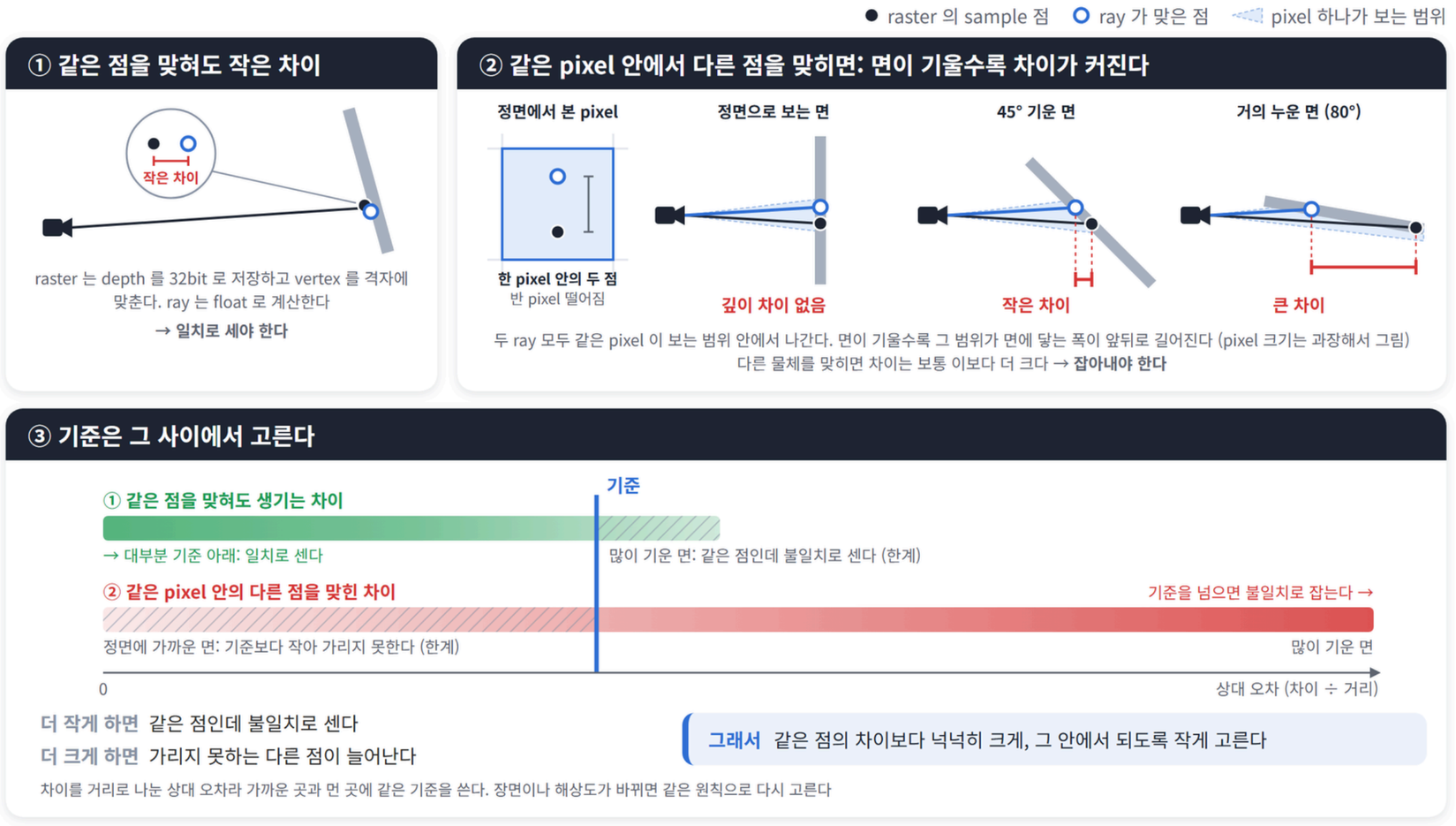
③ pixel 마다 6가지로 분류

둘 다 맞음	Depth Match	Depth Mismatch
한쪽만 맞음	Raster Only Hit	Ray Only Hit
둘 다 안 맞음	Both Miss (하늘)	
alpha test 가 정한 불일치	Alpha Test Mismatch	

→ **heatmap 으로 화면에 칠하고, 화면 전체를 집계**

Rasterizer의 sample 지점과 Ray가 맞힌 hit point가 같은 점인지 검증한다

Infra 검증



이 프로젝트에선 Depth 차이가 1e-4(0.0001) 보다 크면 Depth Mismatch 판정

Infra 검증

Pixels: 2739788 of 2739788

Category	Count	Ratio
Depth Mismatch	228	0.0083%
Raster Only Hit	0	0.0000%
Ray Only Hit	0	0.0000%
Alpha Test Mismatch	9746	0.3557%
Depth Match	2729814	99.6360%
Both Miss	0	0.0000%

Metric	Samples	p50	p99	Max
Depth Error	2587220	5.96e-07 ~ 7.15e-07	6.68e-06 ~ 7.63e-06	1.485

- Depth Mismatch: 0.0083%
- Depth Match: 99.6360%

Infra 검증

Pixels: 2739788 of 2739788

Category	Count	Ratio
Depth Mismatch	27	0.0010%
Raster Only Hit	2	0.0001%
Ray Only Hit	1	0.0000%
Alpha Test Mismatch	0	0.0000%
Depth Match	2404151	87.7495%
Both Miss	335607	12.2494%

Metric	Samples	p50	p99	Max
Depth Error	2404178	4.17e-07 ~ 4.77e-07	3.34e-06 ~ 3.81e-06	0.5395

- Depth Mismatch: 0.0010%
- Both Miss: 12.2494%

Infra 검증

Pixels: 2739788 of 2739788

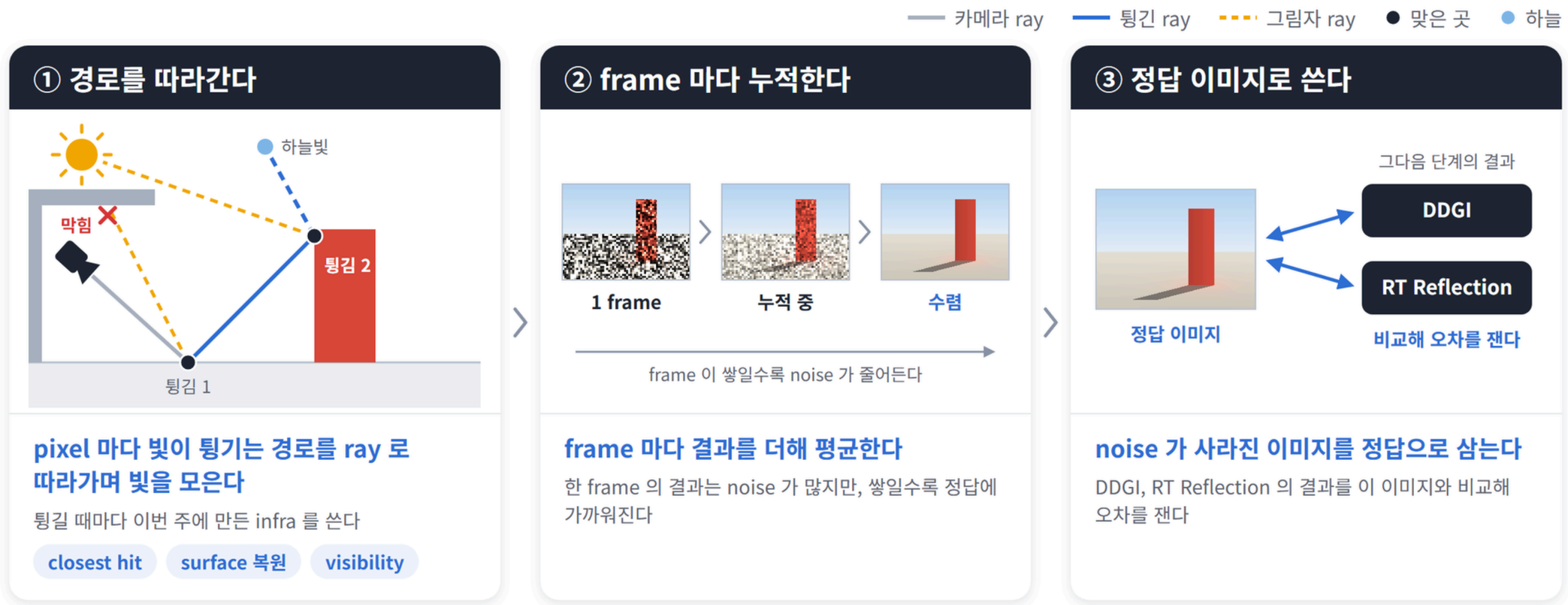
Category	Count	Ratio
Depth Mismatch	125086	4.5655%
Raster Only Hit	0	0.0000%
Ray Only Hit	0	0.0000%
Alpha Test Mismatch	1013	0.0370%
Depth Match	2613689	95.3975%
Both Miss	0	0.0000%

Metric	Samples	p50	p99	Max
Depth Error	2717283	3.81e-06 ~ 4.77e-06	0.000122 ~ 0.000153	32.79

- Depth Mismatch: 4.5655%
- Depth Error: p99에서 허용 오차 살짝 넘는 정도

진행 계획

WEEK 2: Ray Query Infra



Dynamic GI 검증용 Reference Path Tracer를 만든다

thank you!